

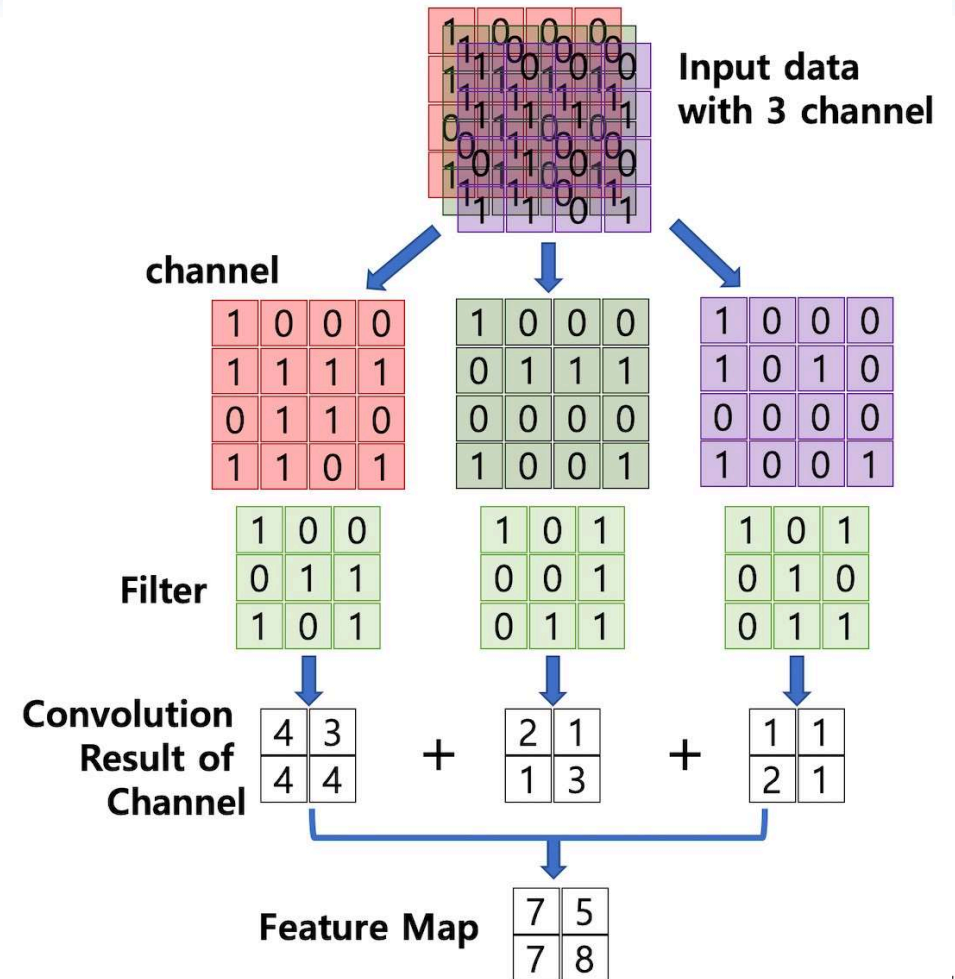
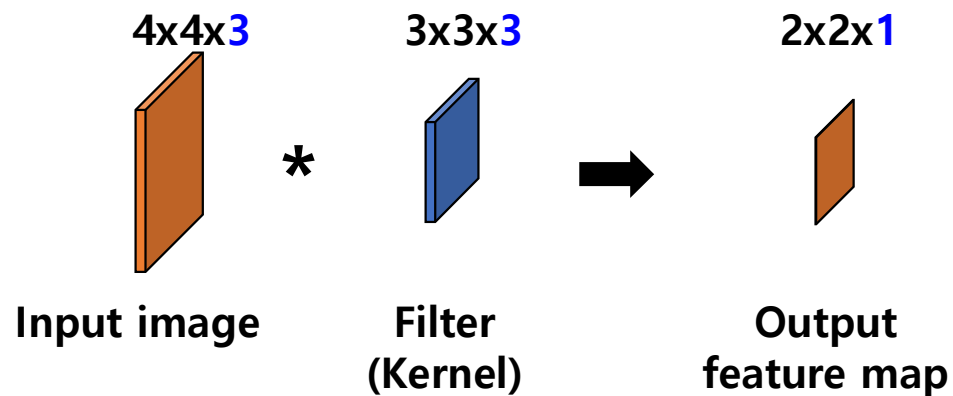
A photograph of four men standing side-by-side. From left to right: Yann LeCun (wearing glasses and a green lanyard), Geoffrey Hinton (wearing a dark jacket), Yoshua Bengio (wearing glasses and a dark jacket), and Andrew Ng (wearing a blue button-down shirt). They are all smiling slightly. The background is a blurred indoor setting.

12주차 이론 (CNN 주요설계모듈)

'인공지능 4대 선구자'로 꼽히는 얀 르쿤, 제프리 힌턴, 요수아 벤지오, 앤드류 응(왼쪽부터). [자료=KAIST]

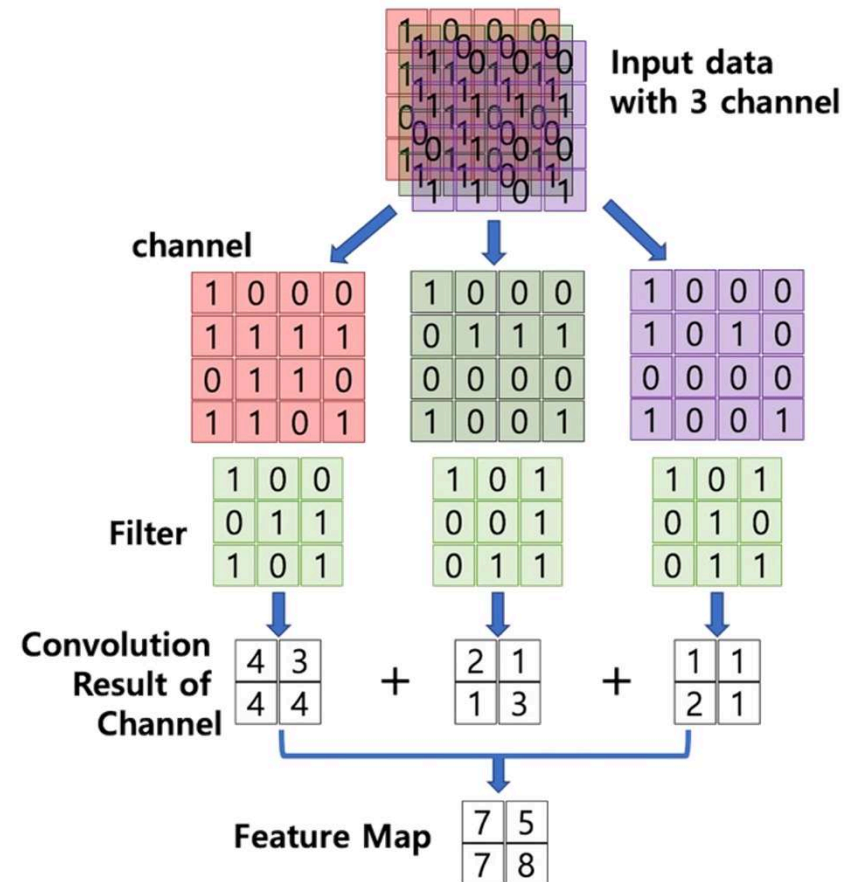
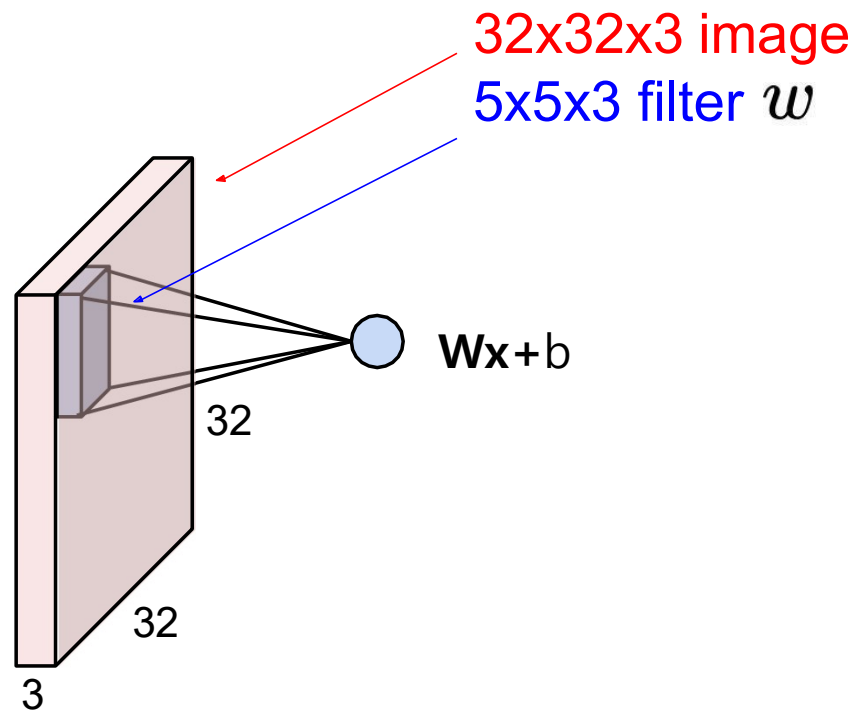
Convolutional Neural Networks - 3D 데이터의 Convolution 연산

- 3D 이미지 (RGB) 입력에 대한 2D convolution 연산



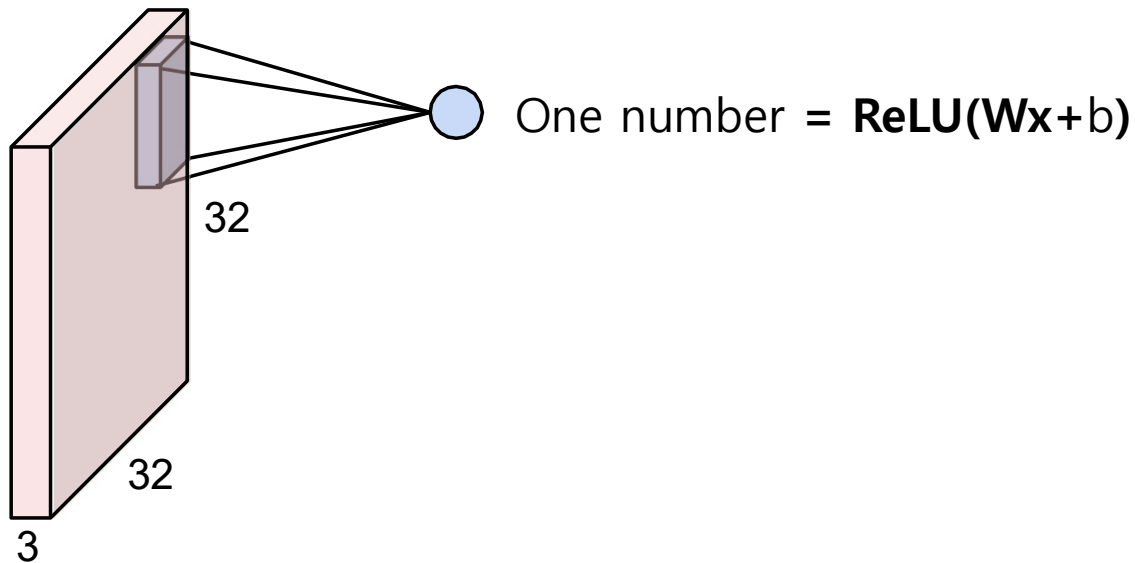
Convolution Layer

Overview



Convolution Layer

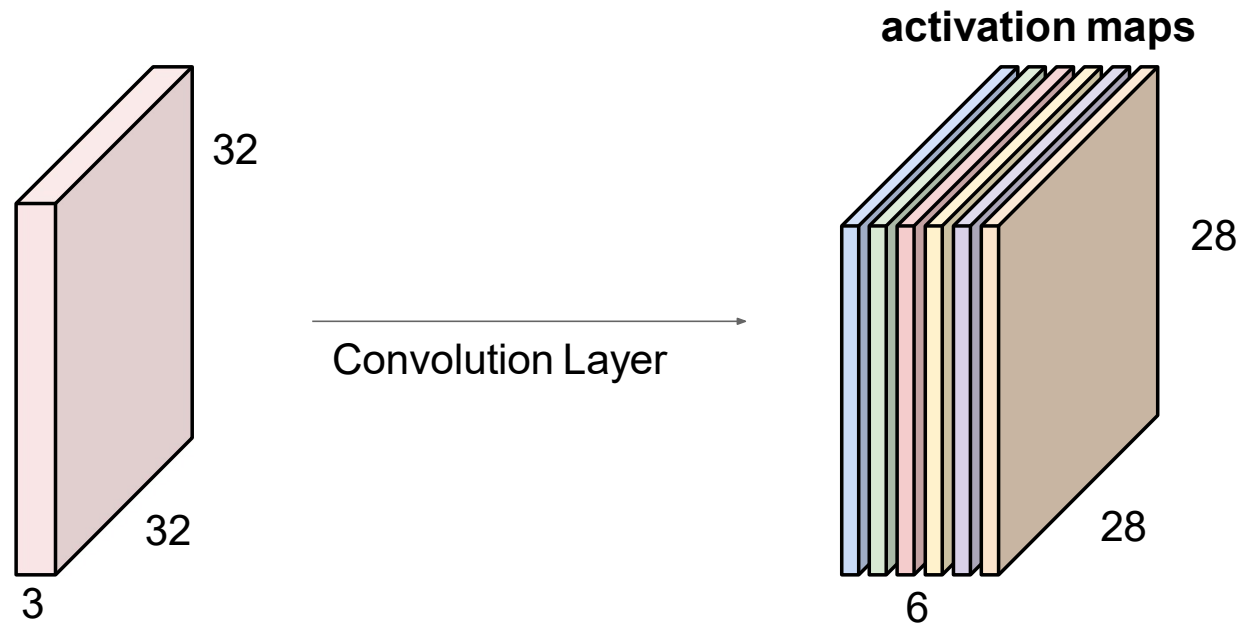
- Overview



Convolution Layer

▪ Overview

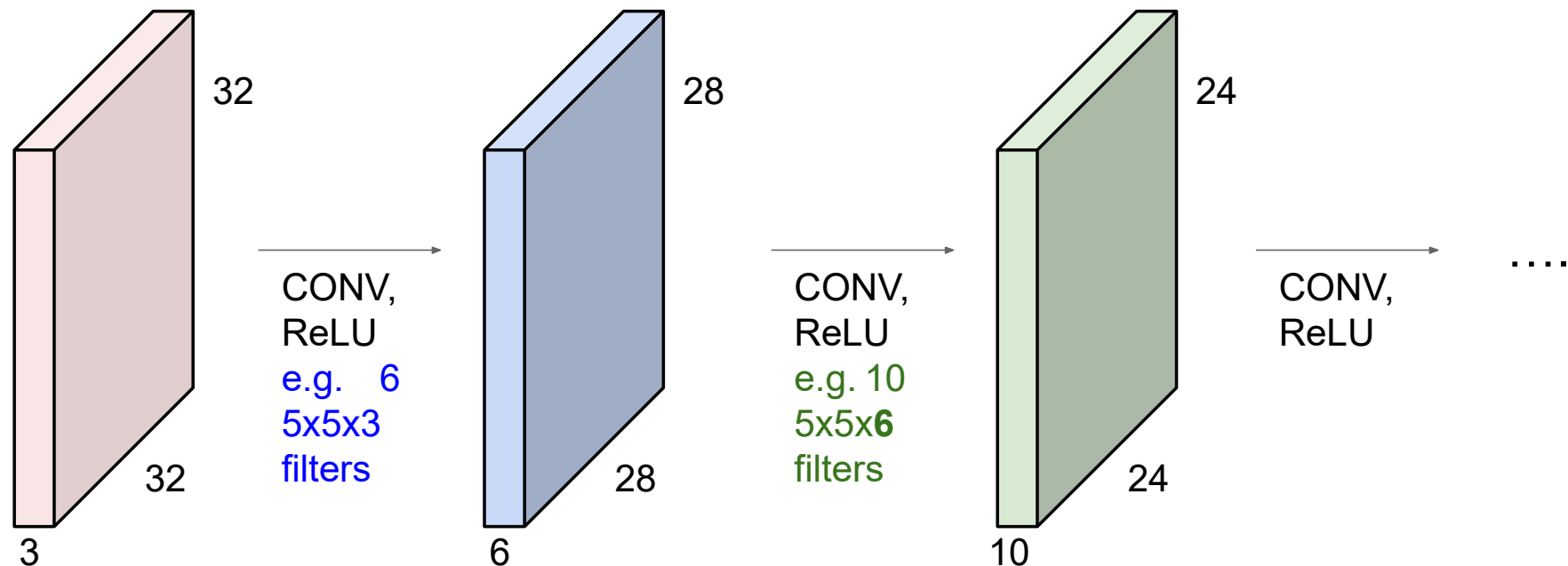
- For example, if we had 6 $5 \times 5 \times 3$ filters, we'll get 6 separate activation maps



Convolution Layer

Overview

- ConvNet is a sequence of Convolutional Layers, interspersed with activation functions

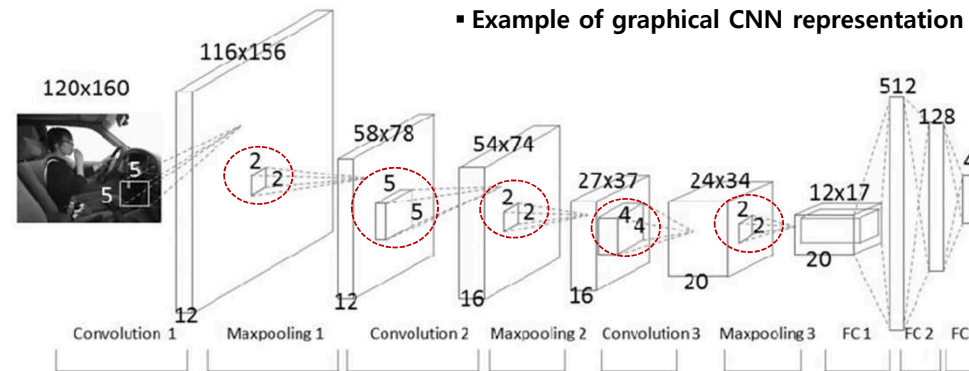


Fei-Fei Li & Justin Johnson & Serena Yeung

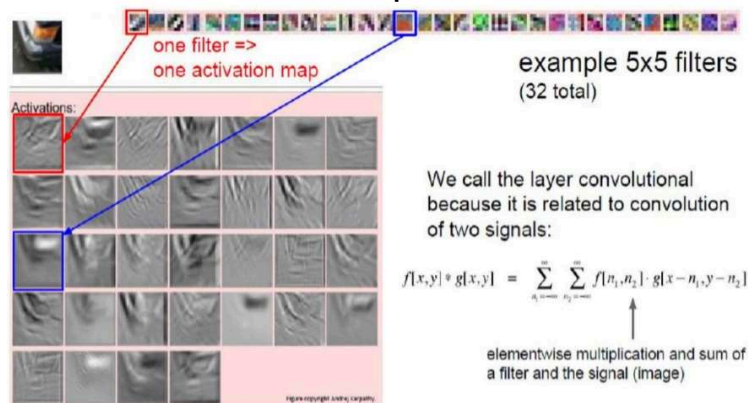
출처: cs231n_2017_lecture5 April 18, 2017

Deep Learning: CNN Visualization

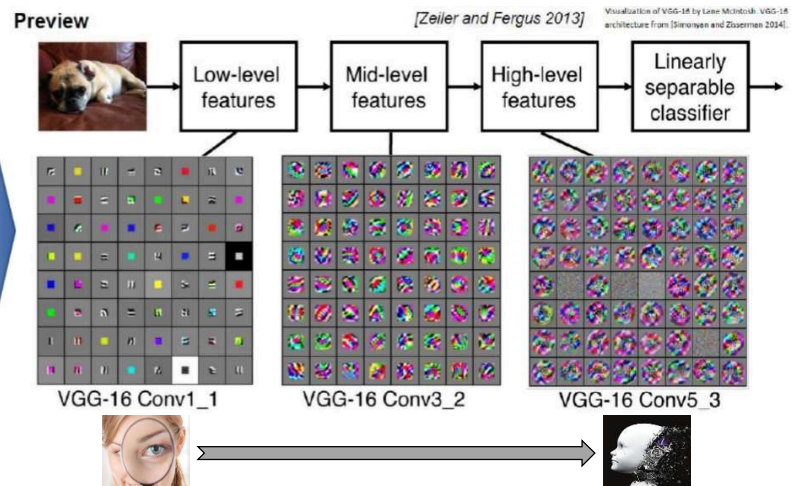
▪ Example of graphical CNN representation



▪ CNN trains the multiple filters (Kernel).

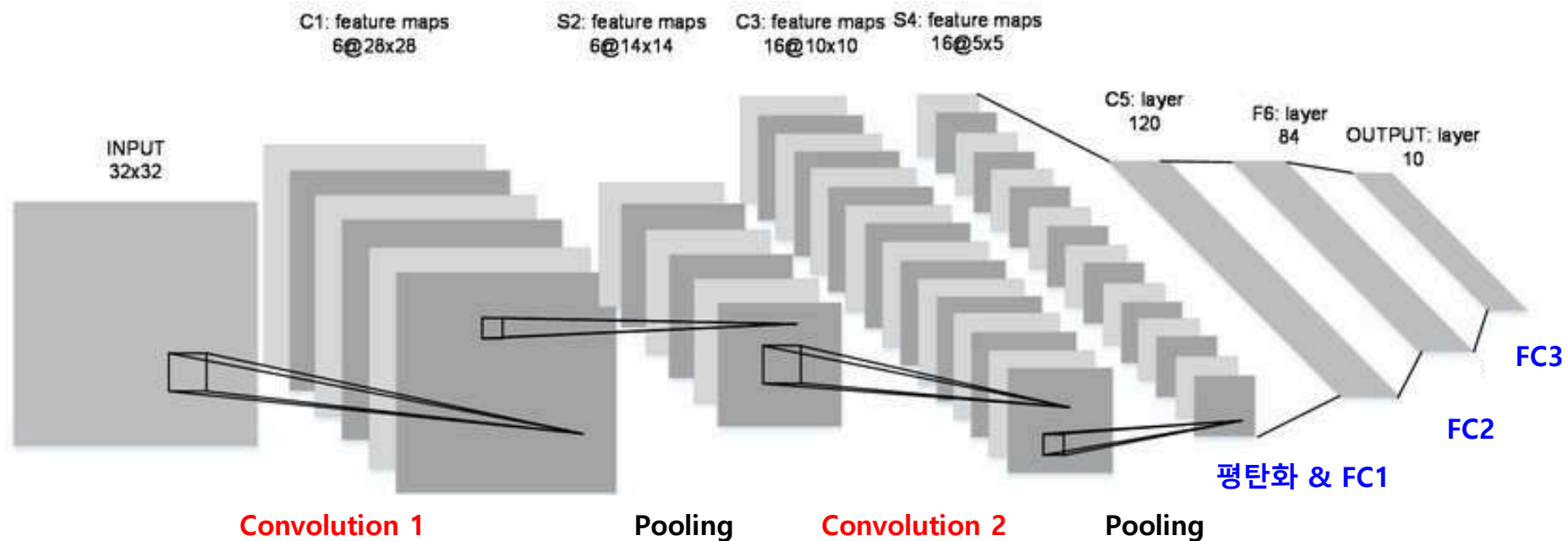


Preview



Convolutional Neural Network (CNN) 이론

- CNN을 이용한 classification model 설계 시 주의사항
 - 일반적으로 CNN의 feature map을 평탄화 한 이후 fully connected layer에 입력함

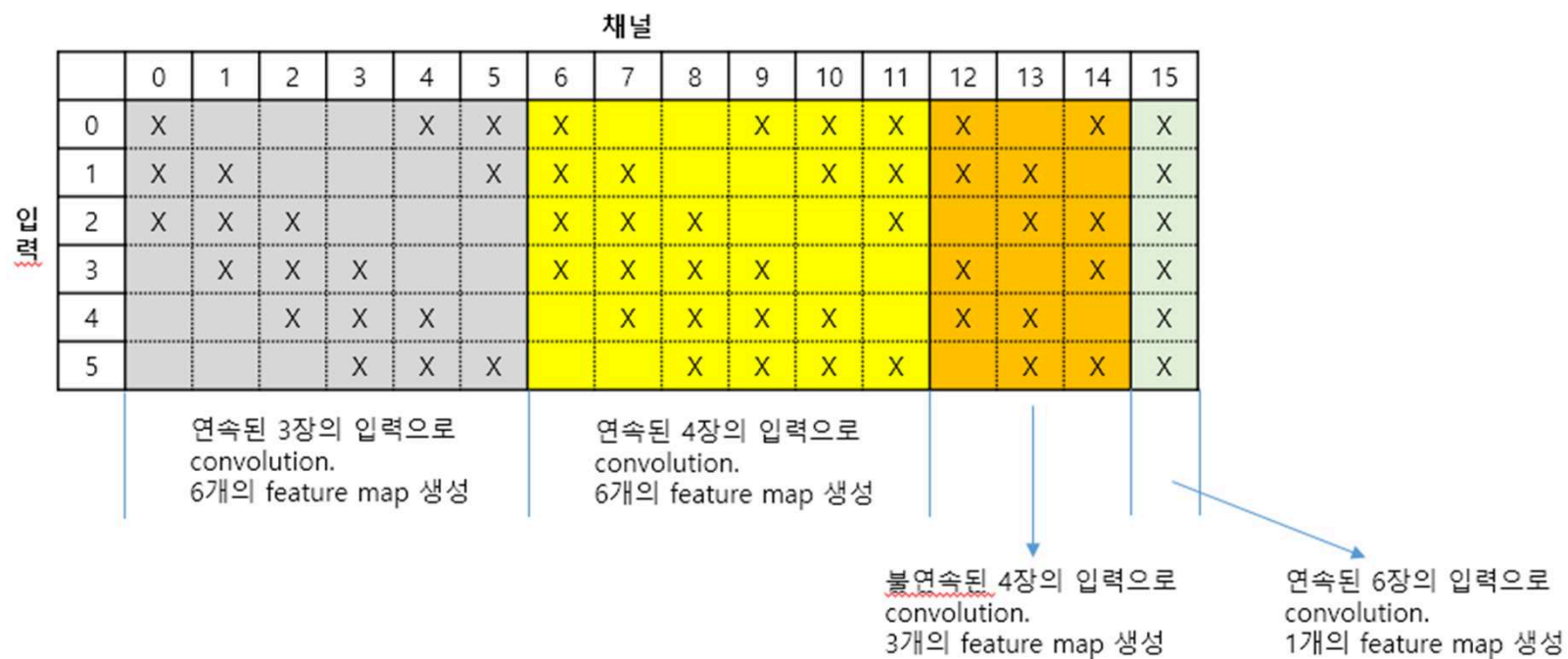


LeNet-5 구조

LeNet-5 구조

- C1** 훈련해야할 파라미터 개수: $(가중치 * 입력맵개수 + 바이어스) * 특성맵개수 = (5 * 5 * 1 + 1) * 6 = 156$
- S2** 훈련해야할 파라미터 개수: $(가중치 + 바이어스) * 특성맵개수 = (1 + 1) * 6 = 12$
- S4** 훈련해야할 파라미터 개수: $(가중치 + 바이어스) * 특성맵개수 = (1 + 1) * 16 = 32$
- C5** 훈련해야할 파라미터 개수: $(가중치 * 입력맵개수 + 바이어스) * 특성맵 개수 = (5 * 5 * 16 + 1) * 120 = 48120$
- F6** 훈련해야할 파라미터 개수: $연결개수 = (입력개수 + 바이어스) * 출력개수 = (120 + 1) * 84 = 10164$

LeNet-5 C3

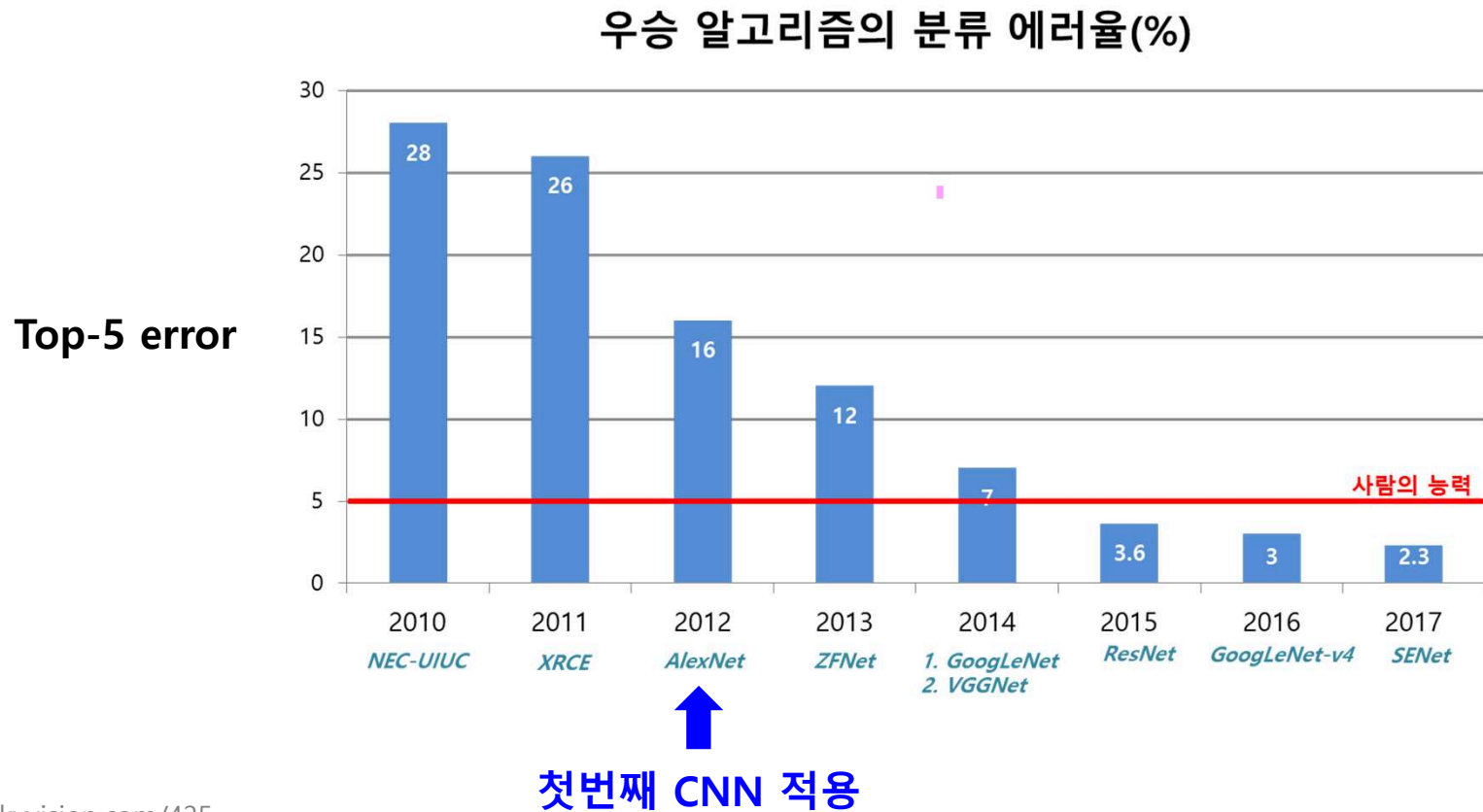


딥러닝 주요모델 구성요소

- Skip connection (ResNet, 2015)
- Dense connection (DenseNet, 2017)
- Channel attention (SENet, 2018)
- Bottleneck Layer
- 구성요소 적용 예시

딥러닝 모델 구성요소

- **ImageNet Large Scale Visual Recognition Challenge (ILSVRC)**
 - 대용량의 이미지셋 (1000개의 클래스) 에 대한 이미지 분류 알고리즘 성능 평가 대회

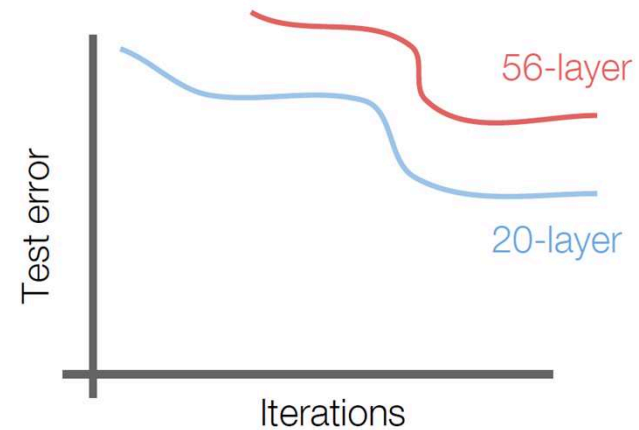
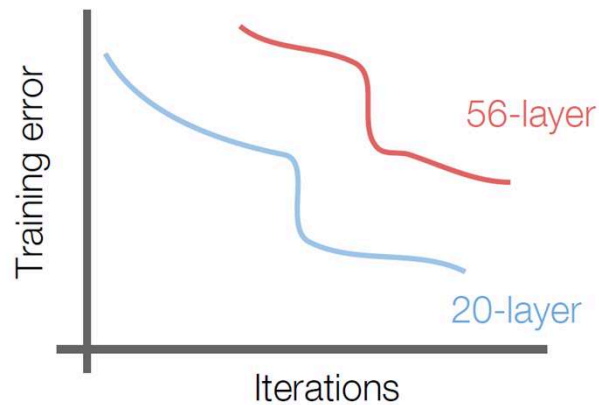


딥러닝 모델 구성요소

- Skip connection (ResNet, 2015)
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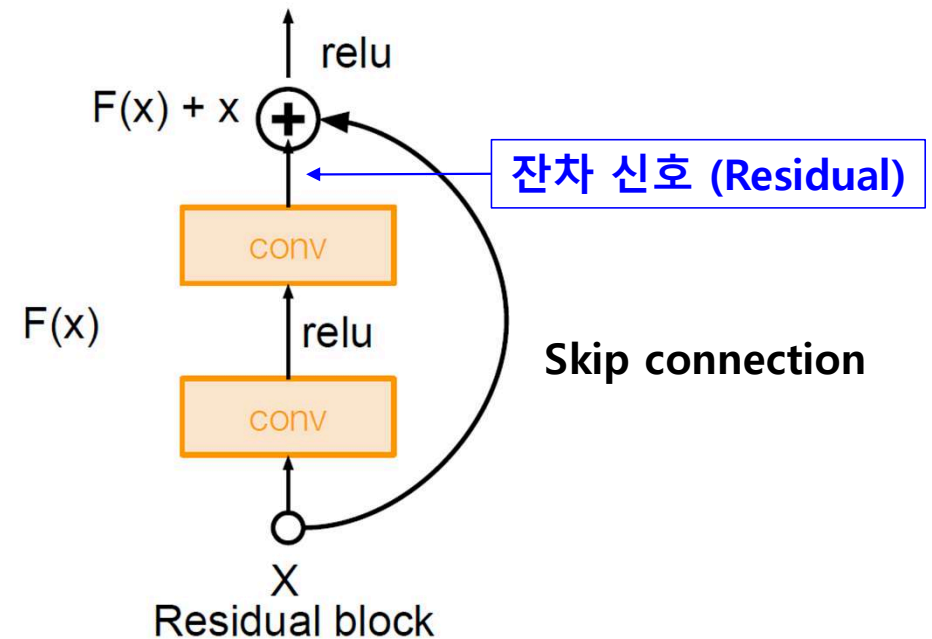
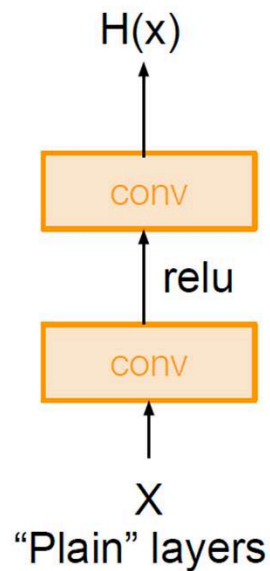
딥러닝 모델 구성요소 - Skip connection

- [CVPR 2015] Deep Residual Learning for Image Recognition (Kaiming He, Microsoft Research)
 - ImageNet dataset에 대해 20-layer, 56-layer 모델의 성능 비교
→ 깊은 모델의 성능이 더 떨어지는 것을 확인



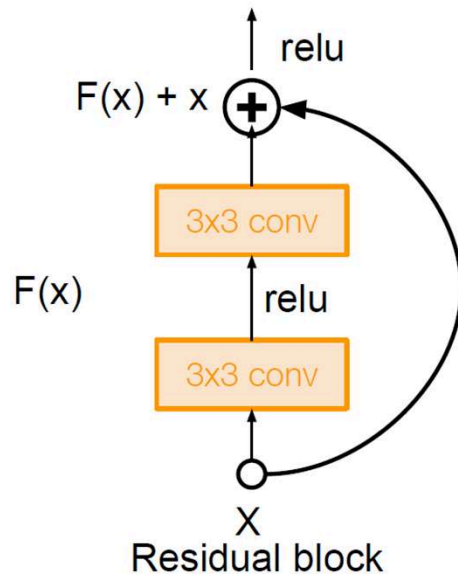
딥러닝 모델 구성요소 - Skip connection

- [CVPR 2015] Deep Residual Learning for Image Recognition (Kaiming He, Microsoft Research)
 - 잔차 신호 (Residual)을 학습 하게 설계 함으로써 문제 해결 시도

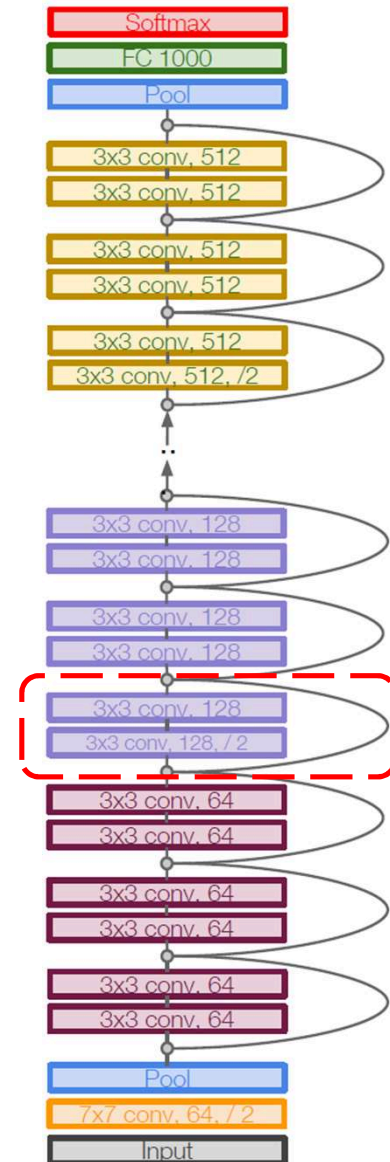


딥러닝 모델 구성요소 - Skip connection

- [CVPR 2015] ResNet (Kaiming He, Microsoft Research)
 - 3x3 convolution 2개,
skip connection으로 구성된 **Residual block** 제안
 - 여러 개의 Residual block을 이용해 제안 기법인 **ResNet**을 구현



Residual block



Inference
진행 순서

딥러닝 모델 구성요소 - Skip connection

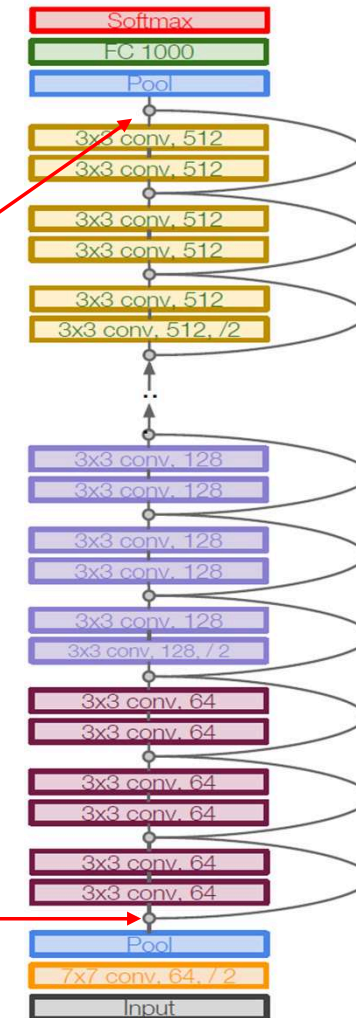
- [CVPR 2015] ResNet (Kaiming He, Microsoft Research)

- Inference가 진행됨에 따라

- feature map의 width, height은 감소됨
 - feature map의 channel은 증가됨

Feature map size: 7x7x512

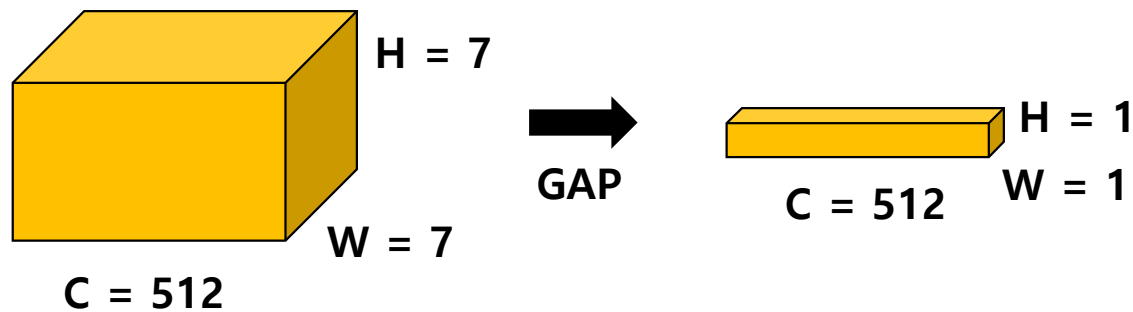
Feature map size: 112x112x64



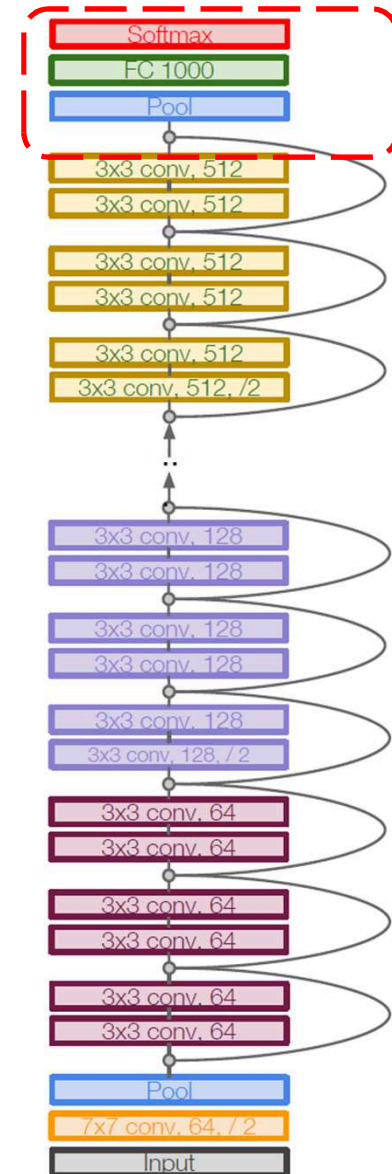
딥러닝 모델 구성요소 - Skip connection

- [CVPR 2015] ResNet (Kaiming He, Microsoft Research)

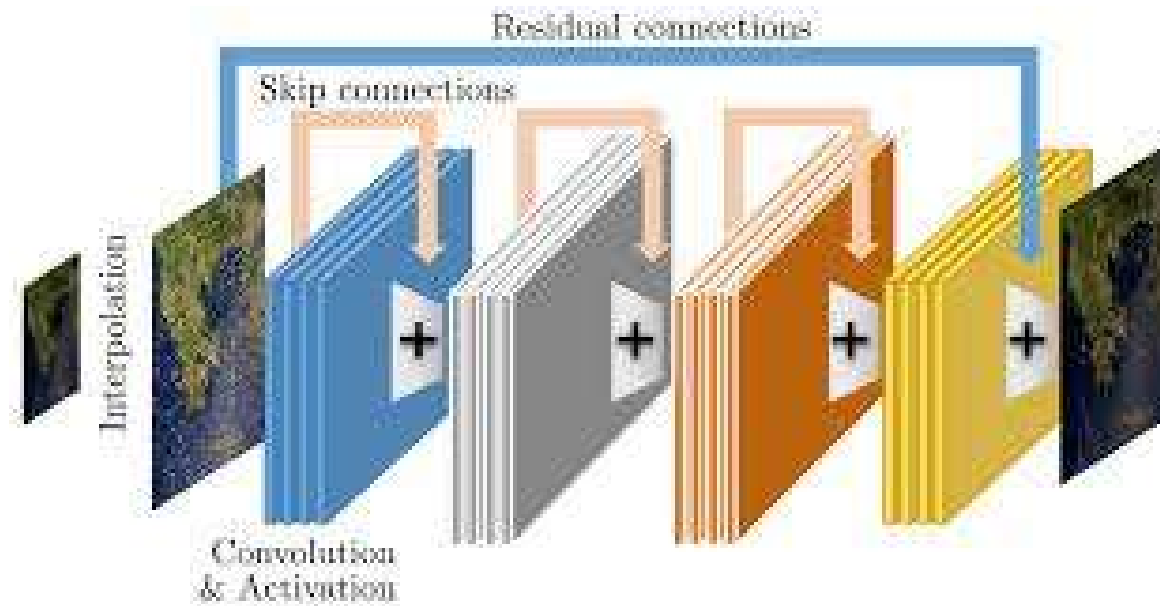
- Convolution layer의 최종 출력은
Global Average Pooling (GAP)을 통해
Fully connected layer에 입력됨



- ❖ C: Channel
- ❖ W: Width
- ❖ H: Height
- ❖ GAP: Global Average Pooling



Skip Connection (ResNet)



딥러닝 모델 구성요소 - Skip connection

- [CVPR 2015] ResNet (Kaiming He, Microsoft Research)
 - 2015년 ImageNet 대회에서는 여러 개의 Layer에 대해 실험한 결과를 제안함

5개의 모델에 대해 실험 결과를 제시

layer name	output size	18-layer	34-layer	50-layer	101-layer	152-layer
conv1	112×112	7×7, 64, stride 2				
		3×3 max pool, stride 2				
conv2_x	56×56	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 64 \\ 3 \times 3, 64 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 64 \\ 3 \times 3, 64 \\ 1 \times 1, 256 \end{bmatrix} \times 3$
conv3_x	28×28	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 128 \\ 3 \times 3, 128 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 4$	$\begin{bmatrix} 1 \times 1, 128 \\ 3 \times 3, 128 \\ 1 \times 1, 512 \end{bmatrix} \times 8$
conv4_x	14×14	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 256 \\ 3 \times 3, 256 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 23$	$\begin{bmatrix} 1 \times 1, 256 \\ 3 \times 3, 256 \\ 1 \times 1, 1024 \end{bmatrix} \times 36$
conv5_x	7×7	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 2$	$\begin{bmatrix} 3 \times 3, 512 \\ 3 \times 3, 512 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$	$\begin{bmatrix} 1 \times 1, 512 \\ 3 \times 3, 512 \\ 1 \times 1, 2048 \end{bmatrix} \times 3$
	1×1	average pool, 1000-d fc, softmax				
FLOPs		1.8×10^9	3.6×10^9	3.8×10^9	7.6×10^9	11.3×10^9

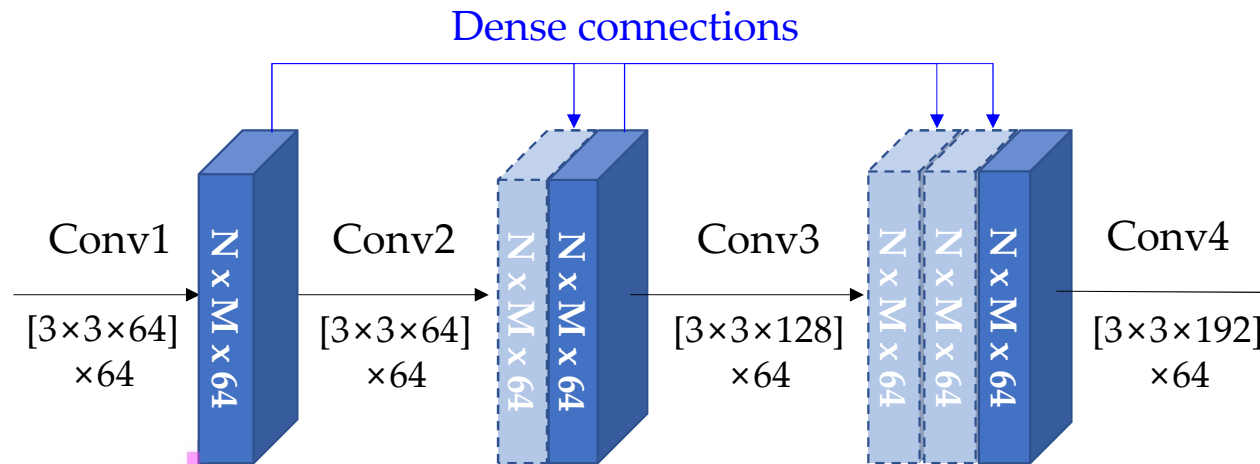
각 모델에 대한 복잡도

딥러닝 모델 구성요소

- Skip connection (ResNet, 2015)
- Dense connection (DenseNet, 2017)
- Channel attention (SENet, 2018)
- Bottleneck Layer
- 구성요소 적용 예시

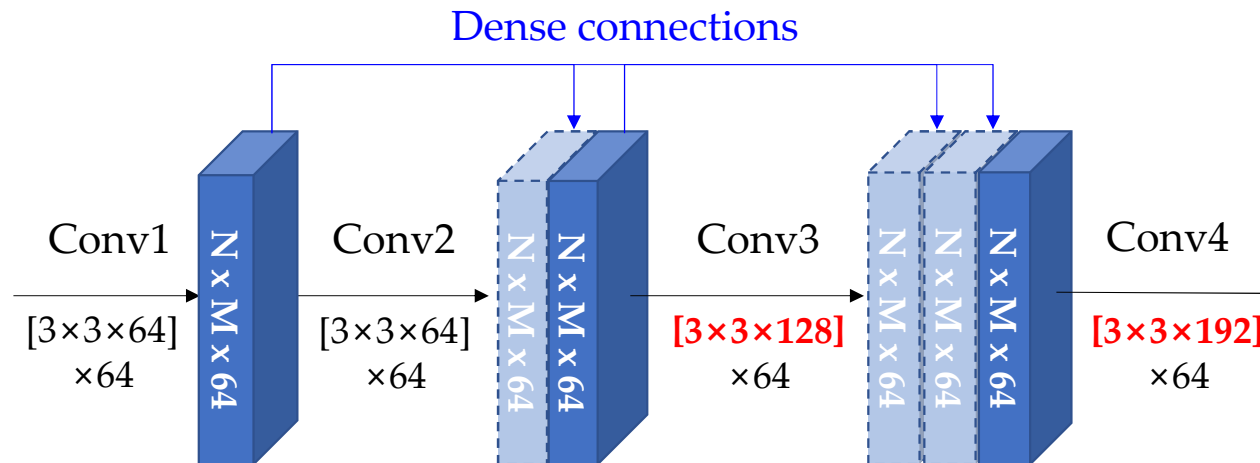
딥러닝 모델 구성요소 - Dense connection

- [CVPR 2017] **Densely Connected Convolutional Networks** (Gao Huang, Cornell University)
 - 이전 Layer의 출력 feature map을 이후 layer에서 재사용



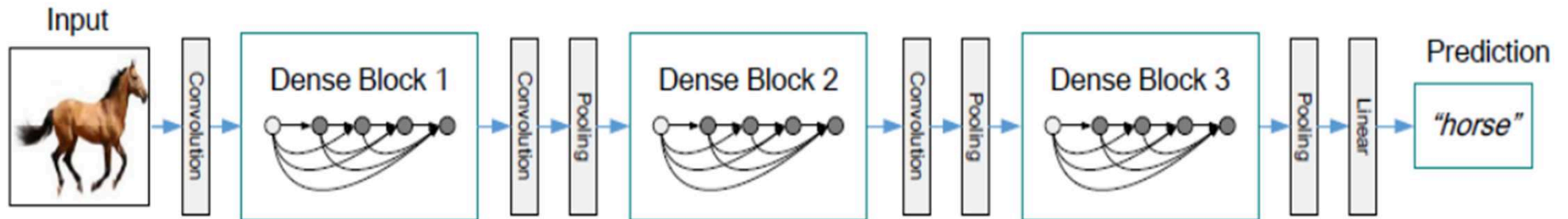
딥러닝 모델 구성요소 - Dense connection

- [CVPR 2017] Densely Connected Convolutional Networks (Gao Huang, Cornell University)
 - 이전 Layer의 출력 feature map을 이후 layer에서 재사용
 - Layer가 깊어짐에 따라 파라미터의 개수가 증가함



딥러닝 모델 구성요소 - Dense connection

- [CVPR 2017] Densely Connected Convolutional Networks (Gao Huang, Cornell University)
 - 여러 개의 Convolution layer를 가지는 Dense Block을 정의,
Dense Block 들로 구성되는 DenseNet을 제안



DenseNet 네트워크 구조 예시

Dense Connection

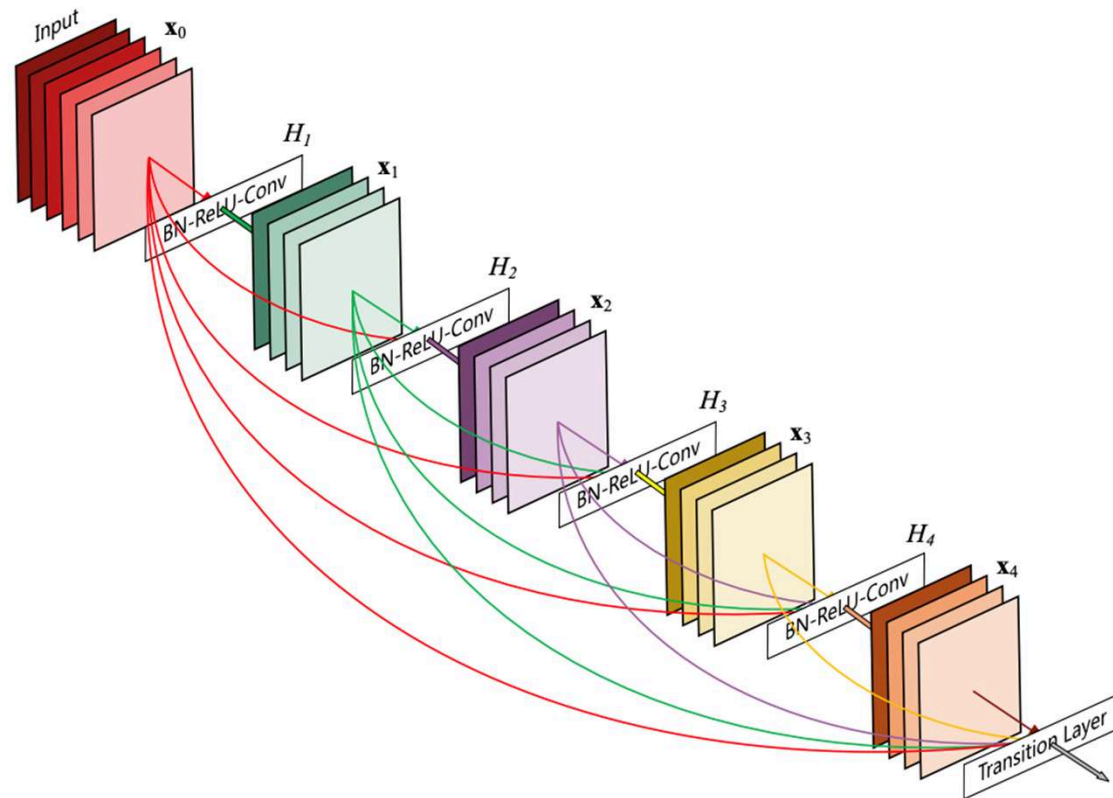


Figure 1: A 5-layer dense block with a growth rate of $k = 4$. Each layer takes all preceding feature-maps as input.

딥러닝 모델 구성요소 - Dense connection

- [CVPR 2017] **Densely Connected Convolutional Networks** (Gao Huang, Cornell University)
 - ResNet보다 깊은 구조에 대해 학습을 진행

5개의 모델에 대해 실험 결과를 제시

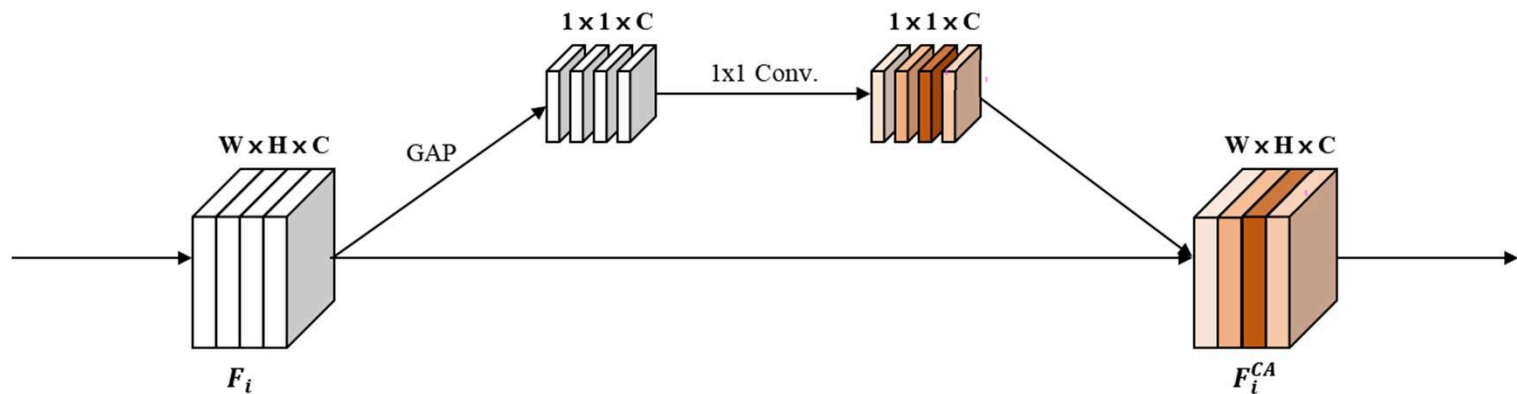
Layers	Output Size	DenseNet-121	DenseNet-169	DenseNet-201	DenseNet-264
Convolution	112×112	7×7 conv, stride 2			
Pooling	56×56	3×3 max pool, stride 2			
Dense Block (1)	56×56	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$
Transition Layer (1)	56×56	1×1 conv			
	28×28	2×2 average pool, stride 2			
Dense Block (2)	28×28	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$
Transition Layer (2)	28×28	1×1 conv			
	14×14	2×2 average pool, stride 2			
Dense Block (3)	14×14	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 24$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 32$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 48$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 64$
Transition Layer (3)	14×14	1×1 conv			
	7×7	2×2 average pool, stride 2			
Dense Block (4)	7×7	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 16$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 32$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 32$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 48$
Classification Layer	1×1	7×7 global average pool			
		1000D fully-connected, softmax			

딥러닝 모델 구성요소

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딥러닝 모델 구성요소 - Channel attention

- [CVPR 2018] Squeeze-and-Excitation Networks
 - Feature map의 각 채널에 대해 중요도를 부여

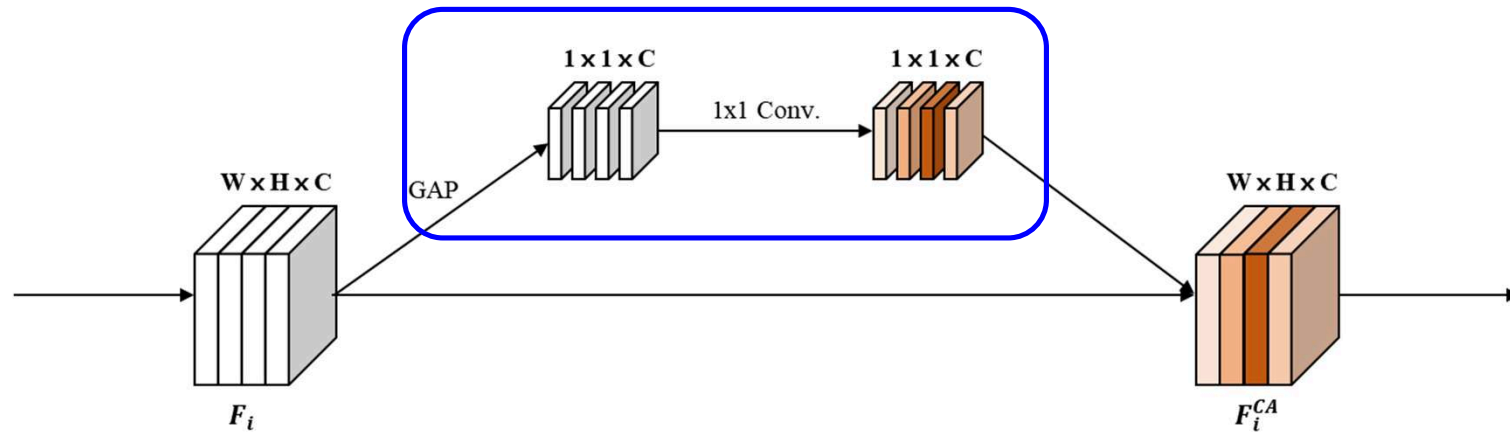


일반적인 feature map

Attention을 통해 중요도가 부여된
feature map

딥러닝 모델 구성요소 - Channel attention

- [CVPR 2018] Squeeze-and-Excitation Networks
 - 일반적으로 GAP를 통해 작아진 feature map에 대해 Fully connected layer 또는 1x1 convolution이 사용됨



딥러닝 모델 구성요소 - Channel attention

- [CVPR 2018] Squeeze-and-Excitation Networks
 - 기존에 제안된 모델들에 대해 Channel attention을 적용하여 결과 제시

	re-implementation			SENet		
	top-1 err.	top-5 err.	GFLOPs	top-1 err.	top-5 err.	GFLOPs
ResNet-50 [13]	24.80	7.48	3.86	23.29 _(1.51)	6.62 _(0.86)	3.87
ResNet-101 [13]	23.17	6.52	7.58	22.38 _(0.79)	6.07 _(0.45)	7.60
ResNet-152 [13]	22.42	6.34	11.30	21.57 _(0.85)	5.73 _(0.61)	11.32
ResNeXt-50 [19]	22.11	5.90	4.24	21.10 _(1.01)	5.49 _(0.41)	4.25
ResNeXt-101 [19]	21.18	5.57	7.99	20.70 _(0.48)	5.01 _(0.56)	8.00
VGG-16 [11]	27.02	8.81	15.47	25.22 _(1.80)	7.70 _(1.11)	15.48
BN-Inception [6]	25.38	7.89	2.03	24.23 _(1.15)	7.14 _(0.75)	2.04
Inception-ResNet-v2 [21]	20.37	5.21	11.75	19.80 _(0.57)	4.79 _(0.42)	11.76

Channel attention을 통한 성능 향상

딥러닝 모델 구성요소 - Channel attention

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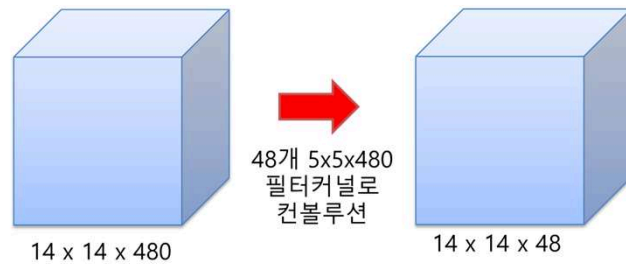
복잡도는 거의 증가되지 않음

딥러닝 모델 구성요소

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- Bottleneck Layer
- 구성요소 적용 예시

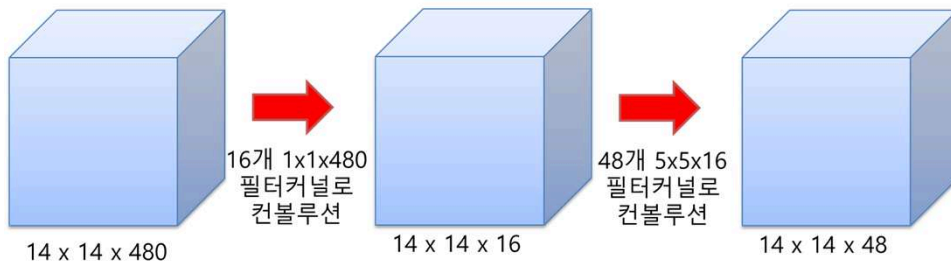
Bottleneck Layer

- 1x1 컨볼루션 사용 → feature map의 개수를 줄이는 목적으로 사용



112.9M
V
5.3M

- ✓ Memory: $[(5 \times 5 \times 480) + 1] \times 48 + (14 \times 14 \times 48)$
- ✓ 연산횟수: $(14 \times 14 \times 48) \times (5 \times 5 \times 480) = \text{약 } 112.9\text{M}$



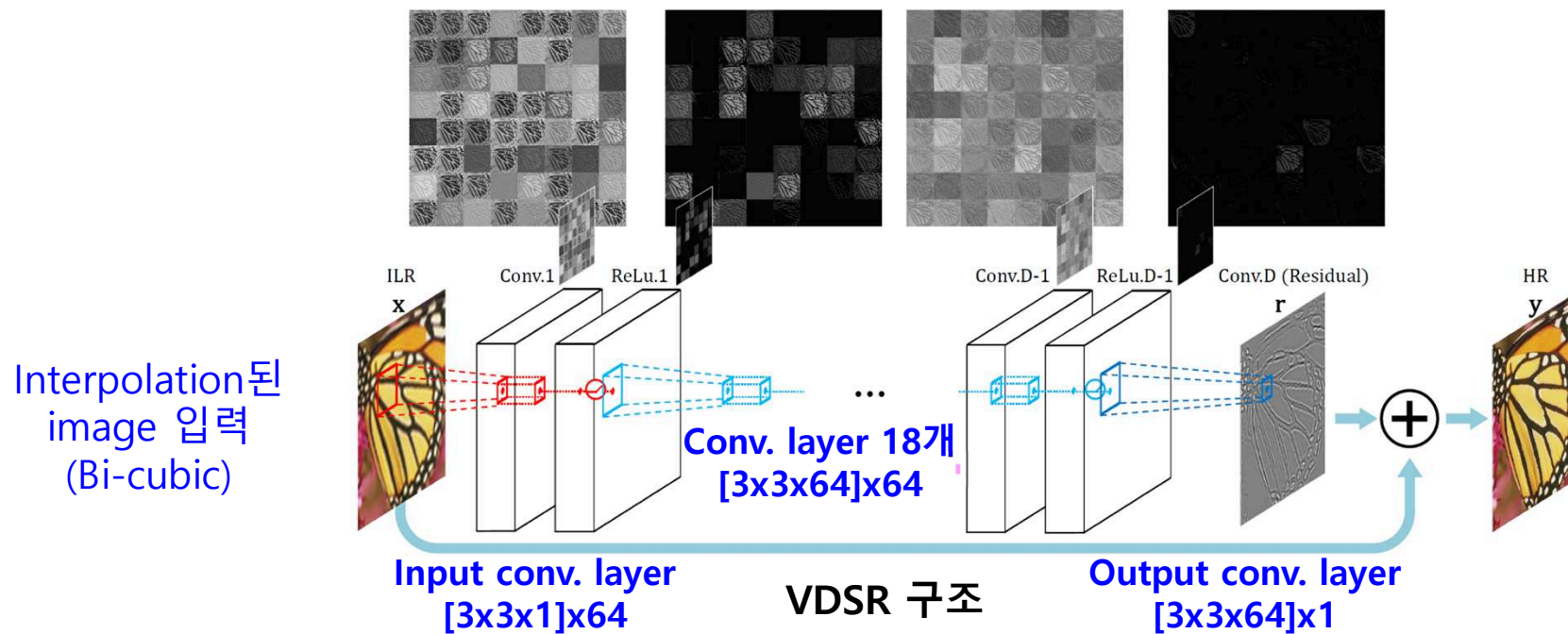
- ✓ Memory: $[(1 \times 1 \times 480) + 1] \times 16 + [(5 \times 5 \times 16) + 1] \times 48 + (14 \times 14 \times 16) + (14 \times 14 \times 48)$
- ✓ 연산횟수: $(14 \times 14 \times 16) \times (1 \times 1 \times 480) + (14 \times 14 \times 48) \times (5 \times 5 \times 16) = \text{약 } 5.3\text{M}$

딥러닝 모델 구성요소

- Skip connection (ResNet, 2015)
- Dense connection (DenseNet, 2017)
- Channel attention (SENet, 2018)
- Bottleneck Layer
- 구성요소 적용 예시

딥러닝 모델 구성요소 - 구성요소 적용 예시

- [CVPR 2016] Accurate Image Super-Resolution Using Very Deep Convolutional Networks (VDSR)
 - 3x3 Convolution 20개 사용, SR 분야에 대해 최초로 Residual Learning 적용



딥러닝 모델 구성요소 - 구성요소 적용 예시

- [CVPR 2017] Image Super-Resolution Using Dense Skip Connections (SR-DenseNet)
 - Dense connection을 이용해 Dense block 8의 출력 단은 1,000개 이상의 feature map을 사용

